Big Data and Machine Learning for Applied Economics

Week 03: Cross-Validation and Regularization: The Honest Pipeline

Eduard F. Martinez Gonzalez

Department of Economics, Universidad Icesi

October 12, 2026

Previously...

- **The line as a prediction machine:** in Cali and its metro area, higher education alone leaves the test RMSE at 6.8 points (the mean, 11.0). Minimizing the RSS *is* minimizing the training error: that is why it flatters (golden rule #1).
- **After predicting:** predicted against observed, residuals as a detector, and the error by quintile and by place, always against a benchmark.
- **In the practice session, the contest of the 511:** every combination of the nine variables, ranked by test RMSE. A five-variable model won, with 6.42.
- **The limitation:** the line does not travel. Country-wide it explains 0.04% of the vote.

Two loose ends

We chose the champion *by looking at the test set*, and to travel the line needs many more columns. Today we tie both: the first with a **judge**, the second with a **knob**.

Today's question: how do we choose a model without fooling ourselves?

Debt 1, from the practice session. The champion of the 511 was chosen and graded with the same test set. Was it the best model, or was it lucky with the 86 stations it happened to get?

Today's question: how do we choose a model without fooling ourselves?

Debt 1, from the practice session. The champion of the 511 was chosen and graded with the same test set. Was it the best model, or was it lucky with the 86 stations it happened to get?

The tool: the judge. Cross-validation estimates the out-of-sample error without leaving the training set: it splits it into pieces, validates on each one, and averages. That is how you compare and choose without touching the test set.

Today's question: how do we choose a model without fooling ourselves?

Debt 1, from the practice session. The champion of the 511 was chosen and graded with the same test set. Was it the best model, or was it lucky with the 86 stations it happened to get?

The tool: the judge. Cross-validation estimates the out-of-sample error without leaving the training set: it splits it into pieces, validates on each one, and averages. That is how you compare and choose without touching the test set.

Debt 2, from "the line does not travel". To predict country-wide, the model needs more variables: the 37 census variables, the department. But every column buys fit with variance: with 37 for 257 stations, OLS already fits the noise.

Today's question: how do we choose a model without fooling ourselves?

Debt 1, from the practice session. The champion of the 511 was chosen and graded with the same test set. Was it the best model, or was it lucky with the 86 stations it happened to get?

The tool: the judge. Cross-validation estimates the out-of-sample error without leaving the training set: it splits it into pieces, validates on each one, and averages. That is how you compare and choose without touching the test set.

Debt 2, from "the line does not travel". To predict country-wide, the model needs more variables: the 37 census variables, the department. But every column buys fit with variance: with 37 for 257 stations, OLS already fits the noise.

The tool: the knob. Regularization charges every coefficient for its size: it shrinks them, lowers the variance, and, with the Lasso, switches off the variables that cannot pay. How much to charge (λ)? That is decided by... the judge.

Today's question: how do we choose a model without fooling ourselves?

Debt 1, from the practice session. The champion of the 511 was chosen and graded with the same test set. Was it the best model, or was it lucky with the 86 stations it happened to get?

The tool: the judge. Cross-validation estimates the out-of-sample error without leaving the training set: it splits it into pieces, validates on each one, and averages. That is how you compare and choose without touching the test set.

Debt 2, from "the line does not travel". To predict country-wide, the model needs more variables: the 37 census variables, the department. But every column buys fit with variance: with 37 for 257 stations, OLS already fits the noise.

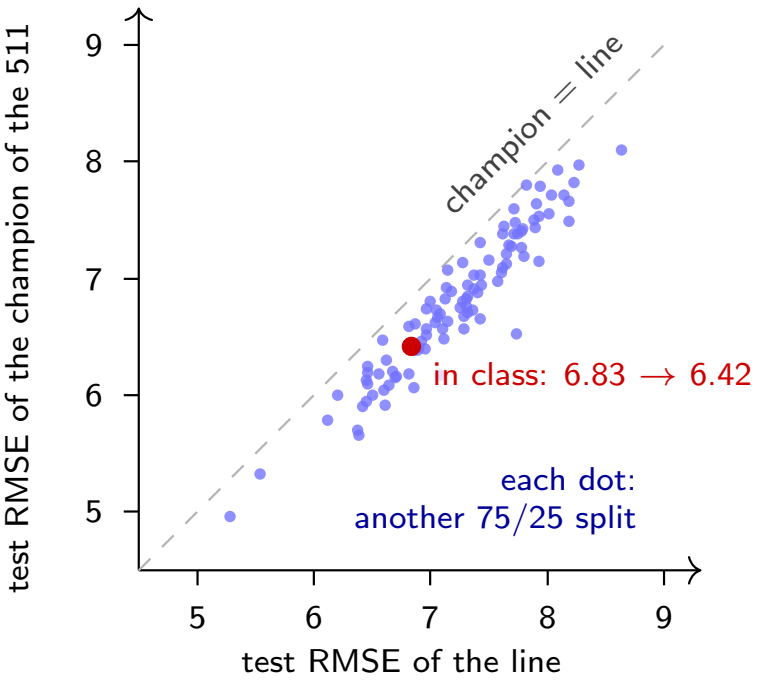
The tool: the knob. Regularization charges every coefficient for its size: it shrinks them, lowers the variance, and, with the Lasso, switches off the variables that cannot pay. How much to charge (λ)? That is decided by... the judge.

The takeaway

They solve different problems, but they need each other: regularization controls complexity and cross-validation decides how much. The test set is opened once, at the end.

The contest of the 511, with one hundred different splits

Same data, same 511 models, same 75/25 rule: the only thing that changes is which stations land in the test set



- **The number depends on the split:** the same line scores from 5.3 to 8.6 depending on which 86 stations it gets.
- **So does the champion:** one hundred contests produced **72 different champions**, with 2 to 9 variables. The one from class won exactly once.
- The champion is never worse than the line: the line is one of the 511 candidates and the champion is whichever erred least on that test set.

What we need

A judge that does not use the test set to choose and does not depend on a single split.

1 The judge: validating without spending the test set

2 The knob: regularization

3 Country-wide

4 Closing

Three roles for the data: learn, choose, and report

In the practice session, the test set played two roles at once: it chose the champion and graded it

In week 2's practice session

training (257 stations) learn: estimate $\hat{\beta}$	**test** (86) choose *and* report

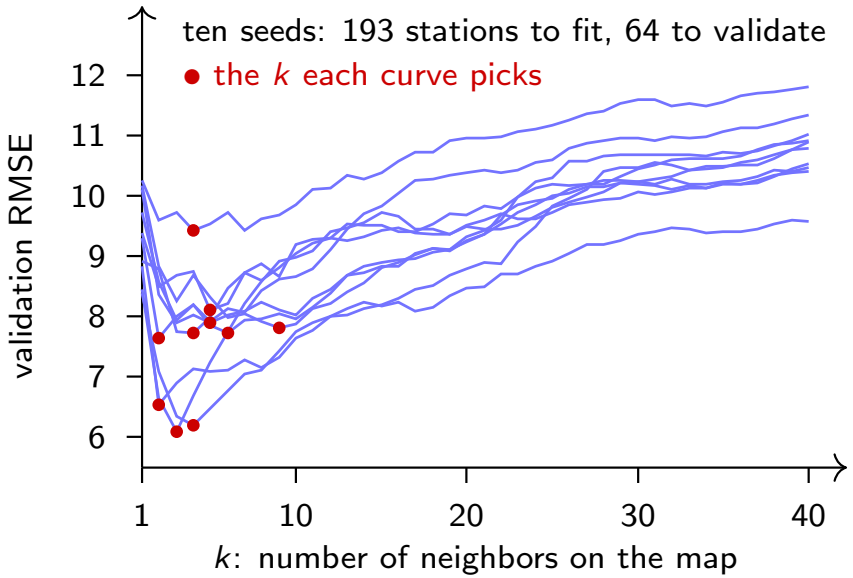
What golden rule #2 demands

training learn: estimate $\hat{\beta}$	**validation** choose	**test** report
	k, variables, λ	exactly once

- **Choosing** means comparing candidates and keeping one; **reporting** means saying how wrong the chosen one will be on new data. They are different questions (ISL, ch. 5).
- **If the same test set does both jobs, the report flatters.** The 6.42 is the smallest of 511 errors, and the smallest of many errors always looks better than it is: $\mathbb{E}[\min_m \widehat{\text{Error}}_m] \leq \min_m \mathbb{E}[\widehat{\text{Error}}_m]$. The more candidates, the more luck gets chosen.
- **The simplest solution** (ISL 5.1.1): separate the roles, with a validation set for choosing and the test set, untouched, for reporting.

A single validation split is a lottery

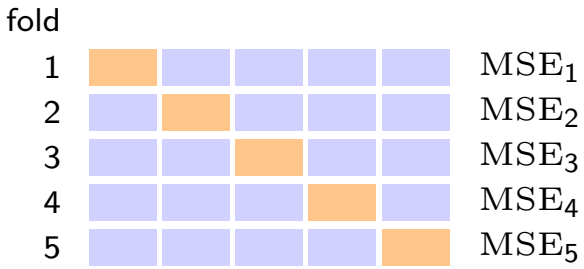
How many map neighbors? Week 2's question, with ten different validation splits



- Each curve picks a different k (from 2 to 9) and sits at a different level: at $k = 10$, from 7.6 to 9.9 depending on the seed. It is ISL's Fig. 5.2 with our stations.
- Two defects (ISL 5.1.1): the result **changes** with the split, and the model is fit with **less data** (193 of 257), so the error tends to overstate.

Cross-validation with J folds: averaging out the lottery

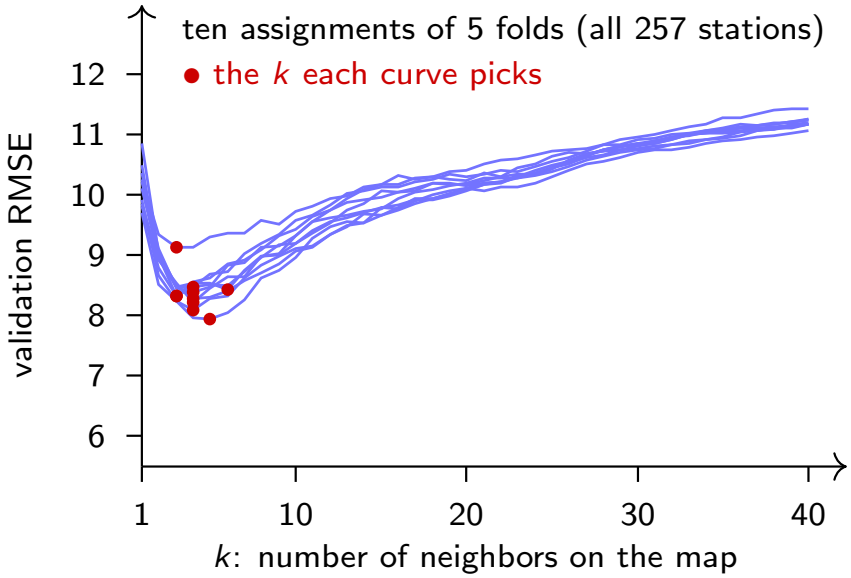
Every station is validated once, by a model that never saw it (ISL 5.1.3)



$$\mathrm{CV}_{(J)} = \frac{1}{J}\sum_{j=1}^{J}\mathrm{MSE}_j$$

$$\mathrm{MSE}_j = \frac{1}{n_j}\sum_{i\in F_j}\left(y_i - \hat{f}^{(-j)}(x_i)\right)^2$$

$\hat{f}^{(-j)}$: the model fit **without** fold j.
We report the root, RMSE_{CV}.



With 5 folds, the ten curves nearly overlap: the chosen k runs from 3 to 6 and the error at $k = 10$, from 9.0 to 9.7 (with a single split: 7.6 to 9.9). It is ISL's Fig. 5.4.

A gift of the method: the J folds give J errors, so the average comes with its own uncertainty, $\mathrm{SE} = \mathrm{s.d.}(\mathrm{MSE}_1, \ldots, \mathrm{MSE}_J)/\sqrt{J}$. (ISL calls it *k-fold CV*; we write J because here k is already taken by the neighbors.)

How many folds? From two to n

Week 2's line on the 257 stations: for each J, fifty different fold assignments

	CV RMSE	s.d.
$J = 2$	7.45	0.14
$J = 5$	7.42	0.05
$J = 10$	7.40	0.03
$J = n$ (LOOCV)	7.40	—
In sample	7.31	

LOOCV (*leave-one-out*) leaves out one station at a time. With OLS there is no need to fit n times (ISL, eq. 5.2):

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_i}\right)^2$$

It is the training residual inflated by $1/(1 - h_i)$, where h_i measures how much y_i weighs in its own fit. The inflation measures how much the line bent toward each station: golden rule #1's optimism, data point by data point.

The judge's own bias and variance (ISL 5.1.4)

With few folds each model learns from less data (the judge overstates the error) and the result dances with the assignment. With $J = n$ there is almost no bias, but you average n nearly identical models and pay for n fits. In practice: $J = 5$ or 10.

The contest of the 511, with an honest judge

The same 511 models, graded by the 10-fold judge inside the training set; the test set no longer chooses: it only reports

Model	Chosen by	Var.	CV RMSE	Test RMSE
The class champion	the same test set	5	7.19	(6.42)
The judge's champion	the CV minimum	4	**7.09**	6.83
The line from week 2	one-standard-error rule	1	7.38	6.83

The contest of the 511, with an honest judge

The same 511 models, graded by the 10-fold judge inside the training set; the test set no longer chooses: it only reports

Model	Chosen by	Var.	CV RMSE	Test RMSE
The class champion	the same test set	5	7.19	(6.42)
The judge's champion	the CV minimum	4	**7.09**	6.83
The line from week 2	one-standard-error rule	1	7.38	6.83

- **The first row's deception:** the 6.42 is in gray because it is not a grade: the same test set that produced it also chose that model among the 511. Its honest grade is the CV, 7.19: 73rd place in the contest.
- **The judge chooses differently:** its minimum is another model (7.09). And the one-standard-error rule finishes the job: among the models within one SE of the minimum —ties, in the judge's eyes—, the simplest is **week 2's line**. Across 20 different fold assignments, it picks the line 19 times.
- **The test set opens at the end, exactly once, to report:** 6.83. Comparing that 6.83 with the (6.42) is comparing a grade with a mirage: one and the same model scores from 5.3 to 8.6 depending on which 86 stations land in the test set (slide 4).

The 7.09 is also a minimum —of 511 CV numbers— and it flatters a little too. Golden rule #2: the number you report comes from the test set, not from the judge.

1 The judge: validating without spending the test set

2 The knob: regularization

3 Country-wide

4 Closing

With many variables, the two usual exits fail

The 37 census variables (education, ages, ethnicity, utilities, strata, households, size), for the same 257 stations

Exit 1: choosing which ones enter (the contest).

- With 9 variables it was 511 models; with 37 it is 2^{37}: **269 million** contests like the one from class. Years of compute.
- And even if you could: the bigger the menu, the more luck weighs. With 511 candidates there were 72 champions; with 2^{37}, the winner would be almost pure noise (ISL 6.1.2).

Exit 2: letting them all in, for free (OLS).

	Coef.	In sample	CV
Line	1	7.31	7.38
OLS with the 37	36	**5.94**	7.17

$R^2 = 0.76$ in sample, with 8 of 36 coefficients significant; but the gap between in-sample and CV multiplies by 18 (from 0.07 to 1.23 points). With 36 coefficients for 257 stations, the model starts to memorize: that is **variance** (ISL, ch. 6).

The third exit: charging admission

In both exits, entering the model is *free*: a variable is either in or out. The third lets all 37 in, but puts a **price** on the size of each coefficient —if you do not contribute, you do not weigh—. The next slide writes that price into the objective function.

Charging for complexity: the penalized objective function

Week 2's sum of squares, plus a price on the size of the coefficients

$$\hat{\beta}(\lambda) \;=\; \arg\min_{\beta} \underbrace{\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j\right)^2}_{\mathrm{RSS}(\beta)\text{: fit}} + \underbrace{\lambda\, P(\beta)}_{\text{the price}}$$

Price $P(\beta)$	Name	What it charges
$\sum_j \mathbb{I}\{\beta_j \neq 0\}$	ℓ_0: the contest of the 511	each variable that enters
$\sum_j \beta_j^2$	ℓ_2: **Ridge**	the size, squared
$\sum_j \lvert\beta_j\rvert$	ℓ_1: **Lasso**	the size

- **The knob λ:** with $\lambda = 0$ it is OLS with all of them; with $\lambda \to \infty$ every β_j is zero and you predict $\bar{y}$, week 2's **benchmark**. In between, a continuum of models for the judge to choose from.
- **Two details that are not details:** the intercept is not penalized, and the x's are **standardized** (ISL, eq. 6.6): a coefficient's price cannot depend on whether a variable is in pesos or in millions.
- Ridge and the Lasso are the tractable versions of the contest: they swap a count that cannot be optimized for a sum that can (ISL, eq. 6.10).

Ridge: shrink everything, switch off nothing

The ℓ_2 price: $\mathrm{RSS} + \lambda \sum_j \beta_j^2$ (Hoerl and Kennard, 1970; ISL 6.2.1)

Ridge divides. In the simple case (orthogonal, standardized predictors; ISL, eq. 6.14):

$$\hat{\beta}_j^R = \frac{\hat{\beta}_j}{1+\lambda}$$

All coefficients shrink by the **same proportion** and none reaches exactly zero.

And the general solution has a formula:

$$\hat{\beta}^R = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{I})^{-1}\mathbf{X}'\mathbf{y}$$

Adding $\lambda\mathbf{I}$ makes $\mathbf{X}'\mathbf{X}$ invertible even with twins or more columns than stations.

What it buys: variance. Dividing stabilizes the coefficients at the cost of a little bias. It pays when OLS is noisy: few observations per variable or highly correlated predictors (ISL, Fig. 6.5).

With week 2's twin: OLS split the credit at random (-0.24 and -0.51, with a standard error 26 times larger); Ridge gives it to both equally. The figure, two slides ahead.

What it does not do: switch off. With 37 columns, the model still has 37. To drop variables you must change the price: **the next slide**.

The Lasso: subtracting instead of dividing

The ℓ_1 price: $\mathrm{RSS} + \lambda \sum_j |\beta_j|$ (Tibshirani, 1996; ISL 6.2.2)

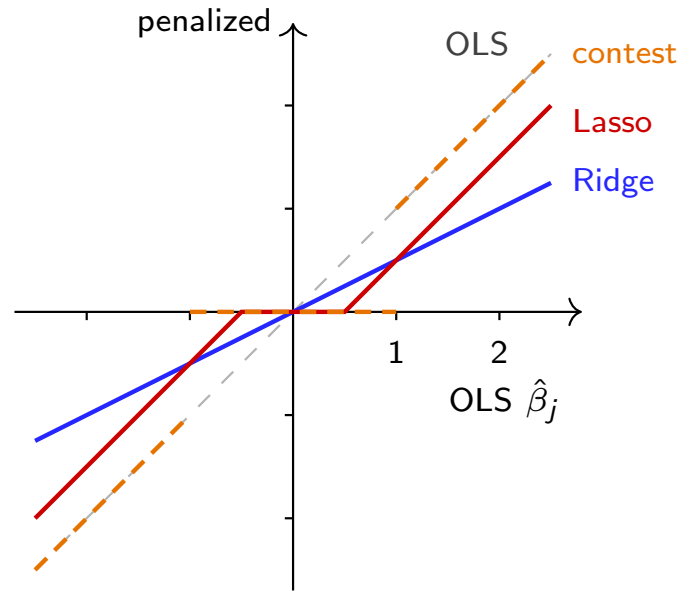
The Lasso subtracts. The same simple case (ISL, eq. 6.15):

$$\hat{\beta}_j^L = \text{sign}(\hat{\beta}_j)\left(|\hat{\beta}_j| - \lambda/2\right)_+$$

It subtracts a fixed amount and, when that is not enough, leaves the coefficient at **exactly zero**: the "soft threshold". (The why, read as marginal cost: appendix A8.)

What it buys: selection. Variables that do not survive the subtraction leave the model; the strong ones stay, shrunken. The final model is shorter: the natural defense when columns get close to rows, as here (37 for 257).

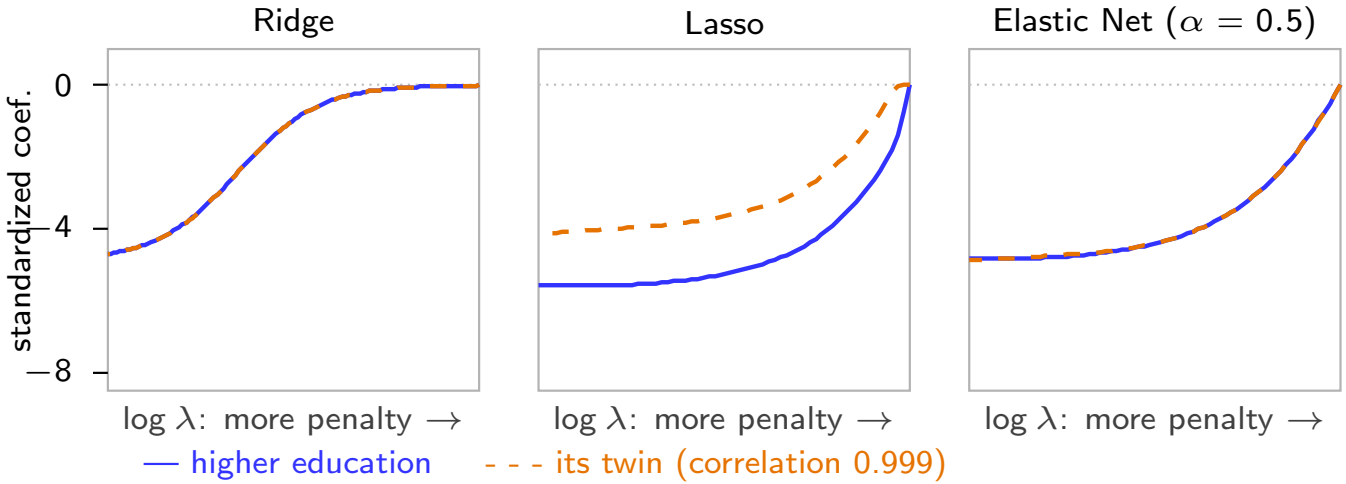
In the figure, the three prices together: Ridge **divides** (previous slide), the Lasso **subtracts**, and the contest **cuts**: all or nothing.



Each curve: what remains of the OLS coefficient under each price, with $\lambda = 1$. Ridge divides by 2; the Lasso subtracts 0.5 and zeroes whatever falls short; the contest erases whatever stays below 1 and leaves the rest untouched (ISL, Fig. 6.10; the contest's cut, ESL, Tab. 3.4).

Elastic Net: both penalties at once

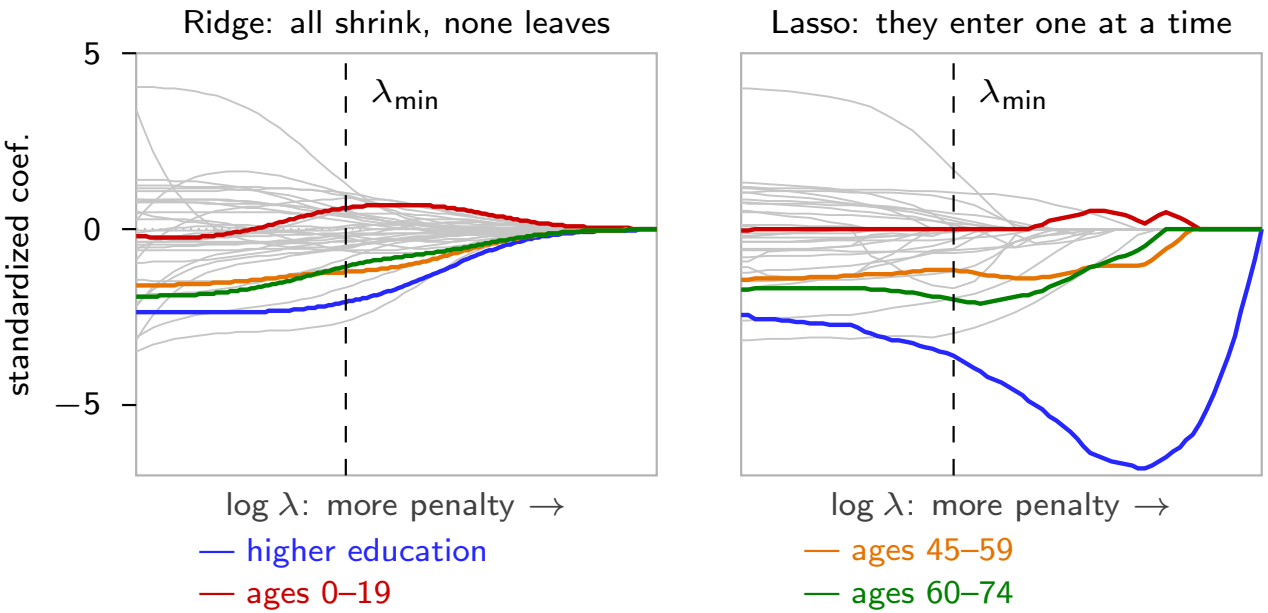
$\mathrm{RSS} + \lambda\left[\,\alpha \sum_j |\beta_j| + \frac{1-\alpha}{2} \sum_j \beta_j^2\,\right]$: with $\alpha = 1$ it is the Lasso and with $\alpha = 0$, Ridge (Zou and Hastie, 2005)



- **Week 2's twin.** Ridge gives both the same weight; the Lasso hands the credit to one and leaves the other behind (a coin flip); Elastic Net treats them like Ridge and, like the Lasso, can switch off what is useless.
- **What it adds:** with **groups** of correlated variables (the census is full of them: strata, education, utilities, ages) it keeps or discards the whole group; and with more columns than observations it is not capped at *n* variables, as the Lasso is.
- Its region sits between the Lasso's diamond and Ridge's circle (the geometry: appendix A9): corners (zeros) and curved edges (grouping). Two knobs, λ and α, both chosen by CV.

The paths: every λ is a model

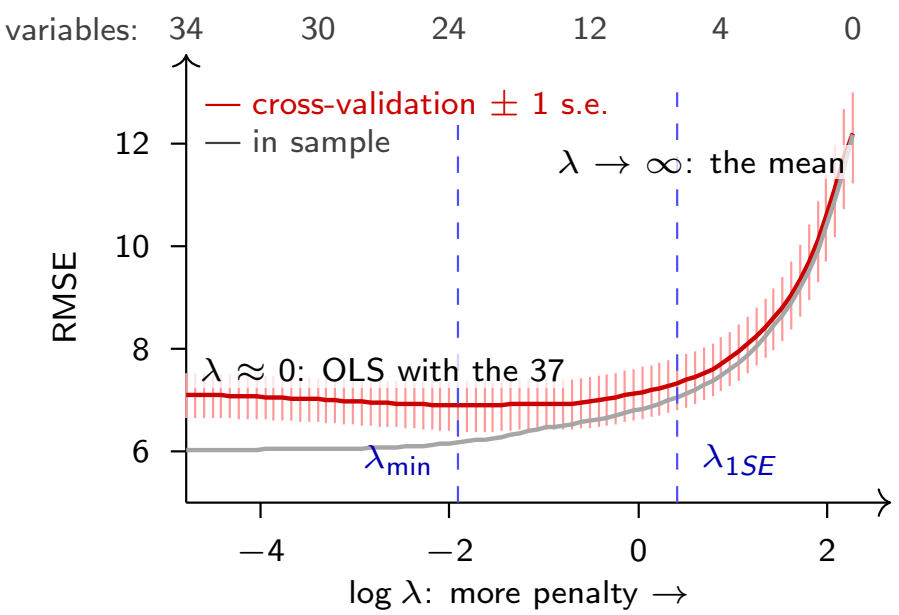
The 37 census variables on the 257 stations: standardized coefficients as the penalty rises (ISL, Figs. 6.4 and 6.6)



- **Ridge:** all shrink together and only reach zero in the limit. **Lasso:** they enter one by one. First, and by far, higher education; then the age structure, stratum 6, and the Afro-descendant share.
- The path is the list of candidates presented to the judge: instead of 2^{37} models, one per value of λ.

How much to shrink? The judge picks λ

10 folds, the minimum, and the one-standard-error rule: the protocol for every knob in the course (ISL 6.2.3)



- **The x axis is the price λ.** Charging nothing ($\lambda \approx 0$): **OLS with the 37 census variables** (CV 7.17). A sky-high price ($\lambda \to \infty$): week 2's **mean**.
- **The two curves:** the gray one (in sample) always rises: the training set would ask you to charge nothing (golden rule #1). The red one (the judge) draws session 1's U.
- **The judge picks:** λ_{min} keeps 24 variables (CV 6.89); λ_{1SE} raises the price and keeps 7 (7.33). **On top:** how many variables survive each price.

The 37 census variables in the metro area: the table

The same 10 folds for everyone; the 86-station test set, opened once

Model	Coef. $\neq 0$	In-sample RMSE	CV RMSE	Test RMSE
Line (higher education)	1	7.31	7.38	6.83
OLS with the 37 census variables	36	5.94	7.17	6.70
Ridge (λ_{min})	37	6.21	6.82	6.51
Lasso (λ_{min})	24	6.17	6.89	6.42
Lasso (λ_{1SE})	7	7.05	7.33	6.95
Elastic Net, $\alpha = 0.5$ (λ_{min})	24	6.20	6.88	6.44

- **Nobody told the model what was expendable: the price found out on its own.** The CV improves from 7.17 (OLS, free admission) to 6.82–6.89. What the contest sought with 2^{37} models, the knob found with a grid of λ.
- **Ridge or Lasso? It is a question about the world, not about the method:** if the signal is spread across many correlated variables, Ridge's dividing works better; if a few carry it, the Lasso's subtracting does (ISL, Figs. 6.8 and 6.9). You cannot know in advance: you ask the judge. Here Ridge wins, barely.
- **With 86 stations the test set crowns no one:** half a point of difference fits inside the noise. Today what matters is the *way* of deciding; the verdict with teeth arrives country-wide.

1 The judge: validating without spending the test set

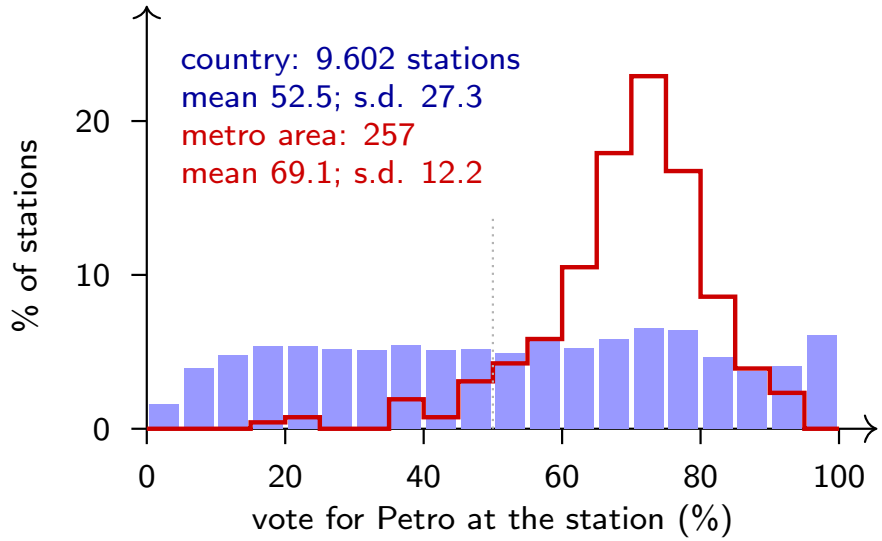
2 The knob: regularization

3 Country-wide

4 Closing

Now the country: the y before any model

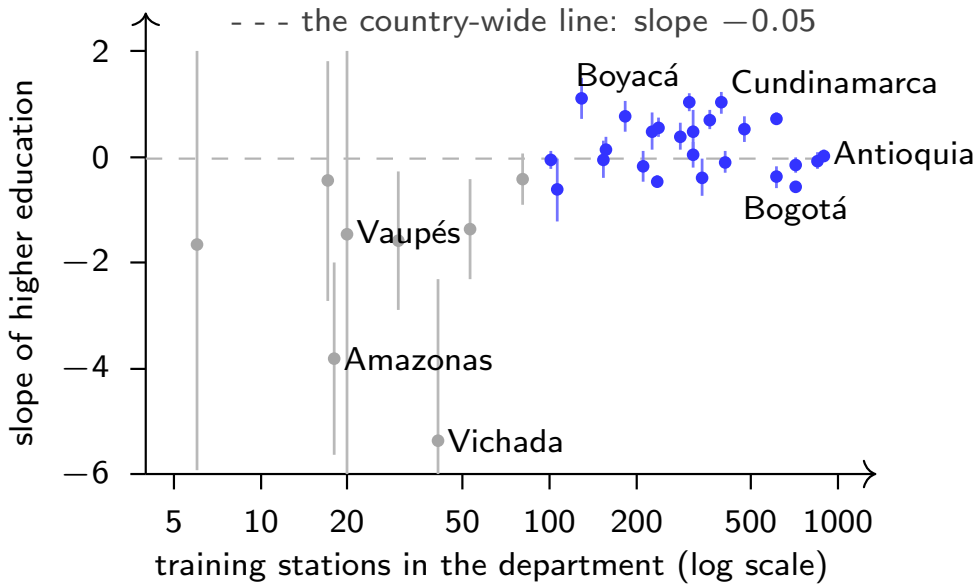
The 9,602 training stations of the course's fixed split (2,398 stay under lock and key)



- Country-wide the y is nearly **flat** from 0 to 100: a standard deviation of 27.3 against 12.2 in the metro area. Petro wins at 53% of the stations.
- Predicting the mean for everyone leaves a test RMSE of 26.7; week 2's line also leaves 26.7: education alone explains 0.04% of the vote.

Education weighs differently in each department

One line per department, with higher education only: its slope ± 2 standard errors, against department size



- **On the right, firm signals with opposite signs:** −0.56 in Bogotá, +1.04 in Boyacá, +1.02 in Cundinamarca. The national line is flat because it **averages opposing relationships**: it explains 0.04 %.
- **The cheap exit:** 33 department dummies: they move the *level*, not the slope (the slopes by hand: $37 \times 33 = 1,221$ extra columns; we will not do it).
- **The funnel is the warning:** on the left the intervals explode (Vaupés: 20 stations): estimating slopes there buys variance; session 5 leaves them to the machine.

Country-wide the dummies pay; the knob no longer does

The same 10 folds for everyone; the 2,398-station test set, opened exactly once

Model	Coef. $\neq 0$	In-sample RMSE	CV RMSE	Test RMSE
Training mean	0	27.3	27.3	26.7
Line (higher education)	1	27.3	27.3	26.7
The 37 census variables	36	20.2	20.3	19.8
Census + department dummies	68	15.6	15.8	**15.6**
with Ridge (λ_{min})	70	15.7	15.8	15.6
with Lasso (λ_{min})	64	15.7	15.8	15.7
with Elastic Net (λ_{min})	65	15.7	15.8	15.7
with Lasso (1-SE rule)	49	15.9	16.0	15.8

- **The department buys 4.2 points** (19.8 $\rightarrow$ 15.6; 95% bootstrap CI, appendix A7: -4.7 to -3.6): place is worth almost as much as the whole census.
- **The knob no longer pays:** the judge pushes λ to the edge of zero and the three penalized models land on OLS's error; the Lasso only switches off 6 twin columns (the logs of housing units and of blocks. . .).
- **Why:** with 9,602 rows for 70 columns there is little variance left to shrink; the knob paid off in the metro area, where 37 columns were chasing 257 stations.

Neighbors that copy each other: what question does your validation answer?

Random folds or whole-municipality folds: week 2's question —how do you validate when neighboring stations look alike?

CV RMSE	Random folds	Whole municipalities	Increase
Training mean	27.3	27.4	+0.1
The 37 census variables	20.3	20.8	+0.5
Census + department dummies	15.8	16.7	+0.9
k-NN on the map ($k = 7$)	**12.0**	**15.0**	+3.0

Neighbors that copy each other: what question does your validation answer?

Random folds or whole-municipality folds: week 2's question —how do you validate when neighboring stations look alike?

CV RMSE	Random folds	Whole municipalities	Increase
Training mean	27.3	27.4	+0.1
The 37 census variables	20.3	20.8	+0.5
Census + department dummies	15.8	16.7	+0.9
k-NN on the map ($k = 7$)	**12.0**	**15.0**	+3.0

- **Two judges, two questions.** Random folds: how well do I predict a new station in a municipality I *know*? Whole municipalities held out: and in one I *never saw*? Neither is "the" error of the model: each answers its own question.
- **The more the model leans on its neighbors, the more it loses in the second:** the map's k-NN loses 3.0 (its only information *is* the neighbor); census + department, 0.9, and the census alone, 0.5. The k-NN still wins, but its lead over the country-wide line shrinks from 3.8 to 1.7.
- **The rule:** the validation scheme mimics the use (Roberts et al., 2017). With data over time: train on one election and test on the next, never at random.

1 The judge: validating without spending the test set

2 The knob: regularization

3 Country-wide

4 Closing

Recap

1. **Choosing is also fitting.** If the test set chooses, it stops being a test: across one hundred splits the contest of the 511 had 72 champions, and the class's 6.42 was the split.
2. **The judge:** CV with $J = 5$ or 10 estimates the out-of-sample error without touching the test set; choose by the minimum or the 1-SE rule, and open the test set once. Judged that way, the contest picks the line.
3. **The knob:** the penalty replaces the per-variable switch with a continuous price. Ridge divides (never switches off); the Lasso subtracts and switches off (corners, flat toll); Elastic Net combines and groups. The judge picks λ.
4. **Together, country-wide:** the department buys 4 points (19.8 $\rightarrow$ 15.6); the knob no longer pays: with 9,602 rows for 70 columns, the judge pushes λ to zero. The knob pays when columns compete with rows, as in the metro area.
5. **The full recipe and the right question:** whatever looks at the y goes inside the folds (golden rule #3), and the folds must mimic the use: random or by municipality is not a technical detail.

Getting ready for next class...

To get the most out of next session

This week's readings:

- ISL, sections 4.1–4.3 (classification and logistic regression) and 4.7.6 (the k-NN-as-classifier lab).
  Guide: section 4.4.2 has the confusion matrix and the ROC curve with the *Default* example; sections 4.4–4.5 (LDA, QDA, Naive Bayes) go to the session's appendix.

Next session — classification: from the logistic to decisions. The same election, a different question: the campaign does not care whether Petro gets 48 or 52 at a station; it cares **whether he wins it**. The y becomes a category; two models for the probability, the LPM and the logistic; and the half that decides: why accuracy deceives, the confusion matrix, the cutoff as a decision with costs, ROC and AUC, calibration.

R application: the country's 12,000 stations, today's variables, and a new y, `gana_petro`. **Problem Set 1 is released** (housing in Cali; due: Tuesday, October 20, 8:00 a.m.).

Essential references

- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2. ed.). Springer. Chs. 5 (5.1) and 6 (6.1, 6.2, and 6.4). [ISL]
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning* (2. ed.). Springer. Secs. 3.4 (Ridge, the Lasso, and Elastic Net) and 7.10 (cross-validation done right and wrong). [ESL]
- Hoerl, A. E., & Kennard, R. W. (1970). Ridge Regression: Biased Estimation for Nonorthogonal Problems. *Technometrics*, 12(1), 55–67.
- Tibshirani, R. (1996). Regression Shrinkage and Selection via the Lasso. *Journal of the Royal Statistical Society, Series B*, 58(1), 267–288.
- Zou, H., & Hastie, T. (2005). Regularization and Variable Selection via the Elastic Net. *Journal of the Royal Statistical Society, Series B*, 67(2), 301–320.
- Friedman, J., Hastie, T., & Tibshirani, R. (2010). Regularization Paths for Generalized Linear Models via Coordinate Descent. *Journal of Statistical Software*, 33(1), 1–22. [`glmnet`]
- Roberts, D. R., et al. (2017). Cross-validation Strategies for Data with Temporal, Spatial, Hierarchical, or Phylogenetic Structure. *Ecography*, 40(8), 913–929.
- Mullainathan, S., & Spiess, J. (2017). Machine Learning: An Applied Econometric Approach. *Journal of Economic Perspectives*, 31(2), 87–106.

5 Appendix: going deeper

A1. Before cross-validation: C_p, AIC, BIC, and adjusted R^2

Correcting the training error instead of estimating the test error (ISL 6.1.3)

With an OLS model of d predictors and $\hat{\sigma}^2$ estimated from the full model:

$$C_p = \frac{1}{n}\left(\mathrm{RSS} + 2d\hat{\sigma}^2\right), \qquad \mathrm{BIC} = \frac{1}{n}\left(\mathrm{RSS} + \log(n)\, d\, \hat{\sigma}^2\right),$$

$$R^2_{\mathrm{adj}} = 1 - \frac{\mathrm{RSS}/(n-d-1)}{\mathrm{TSS}/(n-1)}$$

- They add to the RSS a price per variable: they are, once again, an ℓ_0 price. AIC is proportional to C_p in the Gaussian linear model; BIC charges more ($\log n > 2$ from $n = 8$ on) and picks smaller models.
- They help choose the size in the contest or in stepwise selection (add, or drop, one variable at a time; $1 + p(p+1)/2$ models instead of 2^p, with no guarantee of finding the best).
- Against cross-validation: they assume the correct model and a good $\hat{\sigma}^2$, and they fail as p approaches n (ISL 6.4.2). CV estimates the test error directly, and these days its cost is almost never a problem.

A2. Inside Ridge: it shrinks most where the data inform least

1. The decomposition: with $\mathbf{X} = \mathbf{UDV}'$ (singular values $d_1 \geq \cdots \geq d_p$),

$$\mathbf{X}\hat{\beta}^{OLS} = \sum_j \mathbf{u}_j \mathbf{u}_j' \mathbf{y}, \qquad \mathbf{X}\hat{\beta}^{R} = \sum_j \mathbf{u}_j \frac{d_j^2}{d_j^2 + \lambda} \mathbf{u}_j' \mathbf{y}.$$

2. The reading: Ridge shrinks each direction of the data by the factor $d_j^2/(d_j^2 + \lambda)$. Directions with lots of variation (large d_j) are barely touched; those that barely vary are shrunk nearly to zero.

3. The twin: education and its twin vary together (the "sum" direction, large d_j) and almost never separately (the "difference" direction, tiny d_j). OLS estimates the difference with pure noise (-0.24 versus -0.51); Ridge flattens it and splits the effect equally.

4. Effective degrees of freedom: $\mathrm{df}(\lambda) = \sum_j d_j^2/(d_j^2 + \lambda)$, from p (at $\lambda = 0$) to 0: model complexity as a continuous knob (ESL 3.4.1).

A3. Inside the Lasso: optimality and coordinate descent

1. The optimality conditions (the RSS is not differentiable at zero because of the absolute value; one uses the subgradient): at the solution, for each j,

$$\hat{\beta}_j \neq 0 \;\Rightarrow\; 2\left|\mathbf{x}_j'(\mathbf{y}-\mathbf{X}\hat{\beta})\right| = \lambda, \qquad \hat{\beta}_j = 0 \;\Leftrightarrow\; 2\left|\mathbf{x}_j'(\mathbf{y}-\mathbf{X}\hat{\beta})\right| \leq \lambda.$$

It is the toll from the Lasso slide: a variable enters only if its correlation with the residual, its marginal benefit, reaches λ.

2. Coordinate descent (what `glmnet` does): fix everything but β_j, compute the partial residual $r^{(j)} = \mathbf{y} - \mathbf{X}_{-j}\hat{\beta}_{-j}$, and with $\mathbf{x}_j$ standardized

$$\hat{\beta}_j \leftarrow \text{sign}(z_j)\left(|z_j| - \lambda/2\right)_+ / \,\mathbf{x}_j'\mathbf{x}_j, \qquad z_j = \mathbf{x}_j' r^{(j)}.$$

It is the slide's soft threshold, applied variable by variable until nothing changes.

3. The full path: start at the λ where everything equals zero, $\lambda_{\max} = 2\max_j |\mathbf{x}_j'\mathbf{y}|$, and walk down a grid using each solution as the warm start for the next (Friedman, Hastie, and Tibshirani, 2010).

A4. The Bayesian reading: to penalize is to hold a *prior*

With normal errors and an independent *prior* for each coefficient, $p(\beta) = \prod_j g(\beta_j)$, the posterior mode is

$$\hat{\beta} = \arg\max_{\beta} \left[\log f(\mathbf{y} \mid \mathbf{X}, \beta) + \sum_j \log g(\beta_j) \right]$$

$$= \arg\min_{\beta} \left[\mathrm{RSS}(\beta) - 2\sigma^2 \sum_j \log g(\beta_j) \right].$$

- **Normal** g with mean zero: $-\log g(\beta_j) \propto \beta_j^2$, and the mode is **Ridge** (it is also the posterior mean).
- **Laplace** g (double exponential): $-\log g(\beta_j) \propto |\beta_j|$, and the mode is the **Lasso**. That *prior* has a spike at zero: it expects a priori that many coefficients are exactly zero (ISL, Fig. 6.11).
- λ is the confidence in the *prior*: the larger it is, the more you believe the effects are small. Cross-validation calibrates it with the data.

A5. Nested cross-validation: the error of the choice

- The champion's validation error is the minimum of many: it flatters a little (the three-roles slide). If there is no separate test set —few data, or you want an interval—, nested validation estimates the error of the **whole procedure**, choice included.
- **How:** an outer loop of J folds to evaluate; inside each outer fold, an inner cross-validation loop that picks k or λ using only the inside data. You report the average of the outer errors.
- **What it estimates:** how wrong "choose λ by CV and fit the Lasso" is, not any particular λ. It costs $J_{\text{outer}} \times J_{\text{inner}}$ times more.
- In this course the fixed split plays that role: the 2,398-station test set takes part in no choice.

A6. In R: what is worth knowing about `glmnet`

- **The scale:** `glmnet` minimizes $\frac{1}{2n}\mathrm{RSS} + \lambda\left[\frac{1-\alpha}{2}\sum_j \beta_j^2 + \alpha\sum_j |\beta_j|\right]$. The λ's it reports are not comparable with ISL's, which does not divide the RSS by $2n$.
- **It standardizes internally** (`standardize = TRUE`) and returns the coefficients on each variable's original scale.
- **The same folds for everyone:** `cv.glmnet(x, y, alpha = 1, foldid = pliegue)` lets you compare Ridge, the Lasso, and Elastic Net under the same judge. Then, `lambda.min` and `lambda.1se`.
- **Watch the edge of the grid:** if λ_{min} lands at one end of the grid, the true minimum may lie outside. Country-wide it landed at the *small* end: the judge was asking for $\lambda \to 0$, that is, OLS; you confirm by stretching the grid (`lambda.min.ratio`, `nlambda`).
- **The reference category:** with one, the penalty shrinks toward it. In census + department we use all 33 department columns, so each department shrinks toward the national level.

A7. The bootstrap: manufacturing samples to measure uncertainty

ISL 5.2: resampling with replacement

1. The idea: we have no second sample of the country, but we can manufacture many similar ones: draw n observations **with replacement** from the n we have, recompute the statistic on each resample, and watch how much it varies.

2. How much it leaves out: the probability that an observation never shows up in n draws is

$$\left(1 - \frac{1}{n}\right)^n \longrightarrow e^{-1} \approx 0.368.$$

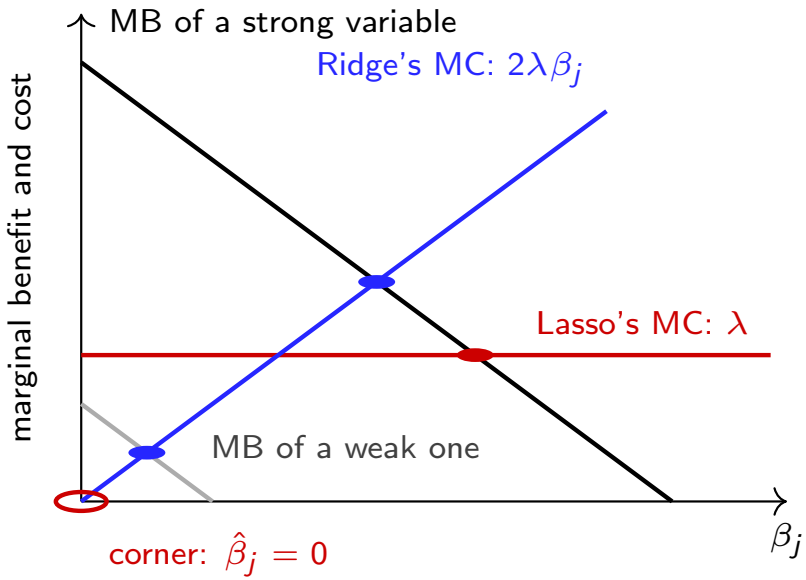
Each resample contains about 63.2% of the distinct observations; with the 9,602 stations, on average 36.8% is left out. That third returns in session 5: the forests' *out-of-bag* error.

3. Today's use: the test set's uncertainty. With 2,000 resamples of the 2,398 test stations, the edge of census + dummies over the 37 census variables alone runs from -4.7 to -3.6 points (95%): the department truly pays, not by luck of the test set.

4. What it does not do: choose models. Resamples repeat stations, so the model sees part of what it is graded on; for choosing, cross-validation.

A8. Why the zero is exact: the price's marginal cost

Read as economists: each variable buys weight as long as its marginal benefit exceeds the marginal cost (the simple case)

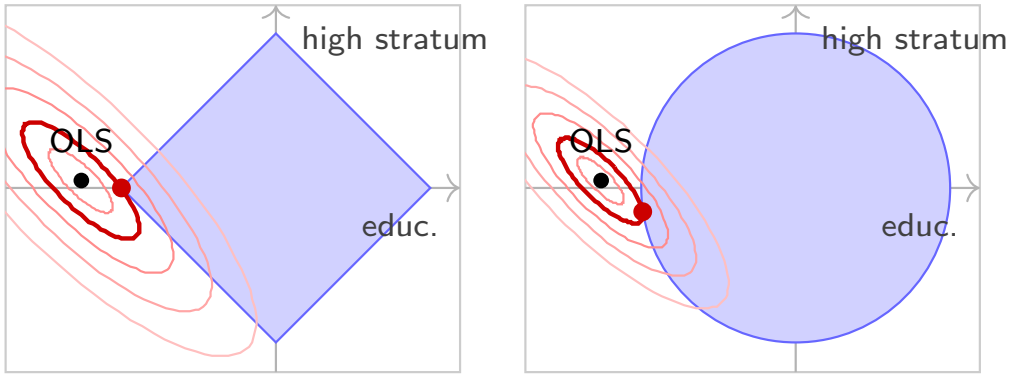


- **The marginal benefit** (black): how much the next unit of β_j lowers the RSS. High at the start, zero on reaching OLS.
- **Ridge** (blue) charges $2\lambda\beta_j$: **the first unit is almost free**, so buying a little always pays. That is why it shrinks and never switches off (blue dots).
- **The Lasso** (red) charges λ **from the very first unit**: a flat entry toll. If the initial benefit does not cover it, the optimum is not to enter: $\hat{\beta}_j = 0$ exactly (the **corner**, red circle).

The zero is not a rounding: it is a corner decision. A variable enters only if its correlation with the residual pays the toll λ (appendix A3).

A9. The geometry: the diamond and the circle

The same problem with a budget: minimize the RSS subject to $\sum_j |\beta_j| \leq s$ (Lasso) or $\sum_j \beta_j^2 \leq s^2$ (Ridge)



Lasso: $(-8.0,\ 0)$ Ridge: $(-7.9,\ -1.2)$

- Higher education and high stratum (the twins from the pairs matrix, correlation 0.81), standardized. Each ellipse joins points with the same RSS around OLS, $(-10.1,\ 0.4)$. Budget: $s = 8$.
- The **diamond** has corners on the axes: the first ellipse to touch it does so at a corner and high stratum lands at **zero**. The **circle** has no corners: Ridge keeps both and passes part of the credit to high stratum. With p variables the diamond has more corners, and there are more zeros (ISL, Fig. 6.7).

A10. A column that already knows the answer (golden rule #3)

Adding the municipality's average vote to the 37 census variables: a natural idea, computed in the wrong place

Census + municipal average	RMSE
CV, average from all data	**11.2**
CV, average within folds	13.1
Test	13.0

A10. A column that already knows the answer (golden rule #3)

Adding the municipality's average vote to the 37 census variables: a natural idea, computed in the wrong place

Census + municipal average	RMSE
CV, average from all data	**11.2**
CV, average within folds	13.1
Test	13.0

- **The idea looks innocent; the error is where it is computed:** with all stations, *before* the fold split. A typical municipality has 5: each station weighs 1/5 in its own predictor. The answer leaked.
- **The deceived judge promises 11.2; the test set charges 13.0.** The variable is not the problem: computed *inside* each fold it is legitimate and even good (13.0: the municipality knows the vote well).

Golden rule #3: cross-validation validates the whole recipe

Whatever learns from the y —picking variables, setting k or λ, group averages— happens **inside** each fold (ISL 6.1.3 demands it even for the contest).